

Abay Zhakenov

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Ph.D. Candidate, Theoretical Condensed Matter Physics
Rutgers University (Advisor: Prof. Natan Andrei)
Expected Ph.D.: 2027

RESEARCH INTERESTS

Quantum impurity and boundary problems in strongly correlated systems; integrability and exact methods (Bethe ansatz, thermodynamic Bethe ansatz, quantum transfer matrix); boundary conformal field theory; non-Hermitian and monitored quantum dynamics; tensor-network methods.

EDUCATION

Ph.D. in Physics

Rutgers University
Condensed Matter Theory Group
Advisor: Natan Andrei

New Brunswick, NJ
Sept. 2021 – Present

M.Sc. (Hons) in Applied Mathematics and Physics

Moscow Institute of Physics and Technology

Sept. 2019 – July 2021
Moscow, Russia

B.Sc. in Applied Mathematics and Physics

Moscow Institute of Physics and Technology

Sept. 2015 – July 2019
Moscow, Russia

RESEARCH EXPERIENCE

Center for Materials Theory, Rutgers University
Graduate Researcher

Piscataway, NJ
2021 – Present

Developed a boundary-adapted thermodynamic Bethe ansatz framework for integrable systems with boundary bound modes, in which the Hilbert space splits into towers built on distinct boundary configurations. The framework resolves a long-standing discrepancy in the impurity entropy of edge impurities and has been applied to one-dimensional superconductors, multichannel Kondo systems, and higher-spin chains.

Current projects: exact-diagonalization and matrix-product study of the quasi-local edge charge labelling the boundary sectors; construction of integrable Kondo models with impurities in antisymmetric higher representations of $su(n)$ by fusion of R -matrices; influence-matrix approach to real-time Kondo dynamics (with the group of J. H. Pixley); non-Hermitian and $\mathcal{PT}$ -symmetric impurity models.

Separately, with Prof. E. A. Yuzbashyan: integrable non-equilibrium dynamics of time-dependent BCS systems and their relation to Knizhnik–Zamolodchikov equations, extending the integrable structure from $SU(2)$ to $SU(3)$ -symmetric pairing models; and a discrete-time path-integral treatment of functional determinants in Migdal–Eliashberg theory.

Landau Institute for Theoretical Physics

Research Assistant, Department of Quantum Field Theory
Integrable bases in superconformal field theory (AGT correspondence)
Advisor: Prof. V. A. Belavin

Chernogolovka, Russia
Oct. 2018 – June 2019

PUBLICATIONS Peer-reviewed

1. P. Kattel, **A. Zhakenov**, and N. Andrei, *Thermodynamics in a split Hilbert space: quantum impurity at the edge of a one-dimensional superconductor*, Phys. Rev. B **113**, 195155 (2026).
2. P. Kattel, **A. Zhakenov**, and N. Andrei, *Multichannel Kondo effect in one-dimensional superconducting leads*, Phys. Rev. B **113**, 165130 (2026).

3. P. Kattel, **A. Zhakenov**, and N. Andrei, *Kondo overscreening in the presence of superconductivity*, Phys. Rev. B **112**, 085103 (2025).
4. P. Kattel, **A. Zhakenov**, and N. Andrei, *Competing color superconductivity and color Kondo effect in quark matter*, Phys. Rev. D **112**, 094012 (2025).
5. P. Kattel, **A. Zhakenov**, P. R. Pasnoori, P. Azaria, and N. Andrei, *Dissipation driven phase transition in the non-Hermitian Kondo model*, Phys. Rev. B **111**, L201106 (2025).
6. V. Belavin and **A. Zhakenov**, *AGT basis in SCFT for $c = 3/2$ and Uglov polynomials*, Nucl. Phys. B **958**, 115133 (2020).

Preprints

1. **A. Zhakenov**, P. Kattel, and N. Andrei, *Thermodynamics in a split Hilbert space: quantum impurity at the edge of the Heisenberg chain*, arXiv:2508.19334.
2. **A. Zhakenov**, P. Kattel, A. Gleis, and N. Andrei, *Boundary phases and thermodynamics of the Kondo spin- s chain: from overscreened Kondo to boundary-bound states*, arXiv:2608.12453.
3. P. Kattel, **A. Zhakenov**, and N. Andrei, *Breakdown of monotonic impurity entropy flow in $\mathcal{PT}$ -symmetric multichannel Kondo systems*, arXiv:2608.04083.
4. P. Kattel, **A. Zhakenov**, and N. Andrei, *Monotonic impurity entropy beyond unitarity: the $\mathcal{PT}$ -symmetric quantum impurity model*, arXiv:2606.28495.

AWARDS

Peter Lindenfeld Graduate Fellowship for excellence in academics, research, and teaching	Rutgers University, 2026
Louis Bevier Dissertation Completion Fellowship (alternate)	Rutgers University, 2026
Langreth Graduate Development Award	Rutgers University, 2023
Boyd Scholarship	Rutgers University, 2022
Scientific Leadership Support Project Scholarship	MIPT, 2019–2020
Third Prize, TUYMAADA International Physics Olympiad	2014
Second Prize, Kazakhstan Republican Physics Olympiad	2014

TALKS & PRESENTATIONS

- **Dynamics and thermodynamics of quantum impurities at the edge of spin chains**, APS Global Physics Summit (2026), Denver.
- **Impurity effects in the spin- S Heisenberg–Babujian chain**, APS Global Physics Summit (2025), Anaheim.
- **Thermodynamics in a split Hilbert space: quantum impurity at the edge of 1D systems**, Rutgers Fall CMT Symposium (2025); Seminar on Nonperturbative Methods in QFT (2025).
- **No man is an island, and so the quantum impurity isn’t...**, SSPAR Graduate Seminar (2025).
- **Integrability and time-dependent Hamiltonians**, NJ Quantum Matter & Information Forum (2022).
- **Derivation of the Painlevé equations for two-spin correlation functions in the Ising model**, Seminar, School of Mathematical Physics (2019).
- **Poster: Impurity effects in the spin- S Heisenberg–Babujian chain**, Rutgers Spring CMT Symposium (2025).

**TEACHING &
MENTORING**

Lecturer, Honors Physics III (PHY 273), Rutgers
Sole lecturer for a full-semester honors course.

Fall 2026

Instructor & Course Organizer,

Summer 2025

Quantum Computation: Foundations and Applications

Designed and led a six-week intensive course for high-school students on quantum algorithms, qubit dynamics, and Qiskit laboratory work. Mentored seven students through research projects on variational quantum algorithms, quantum walks, and quantum classifiers, culminating in a symposium reviewed by Rutgers faculty.

Instructor & Course Organizer,

Summer 2026, 3 seasons

Fundamentals of Modern Physics

Founded and taught a 15-hour lecture course on quantum mechanics, quantum information, condensed matter, and particle physics for high-school students. Students carried out independent projects alongside the lectures, recent examples including a simulation study of quantum versus pseudo-random number generation, a numerical derivation and test of the quantum bit error rate that exposes an eavesdropper in the BB84 protocol, and an interactive three-body dynamics simulator.

Lecturer, *Engineers of the Future* Program, Rutgers

Summers 2025, 2026

Rutgers School of Engineering program for under-resourced New Jersey students. Designed and taught a foundational physics course preparing them for engineering studies at Rutgers.

Teaching Assistant, Rutgers University

2021 – Present

Recitations, laboratories, and occasional lectures across eleven undergraduate physics courses (PHY 115, 116, 202, 203, 204, 205, 229, 273, 289, 301, 324; grading for 385, 397, 501).

Moscow Institute of Physics and Technology

2019 – 2021

Teaching assistant, *Introduction to Quantum Field Theory (Advanced)*; lecturer on random partitions in *Selected Topics of Theoretical and Mathematical Physics*; seminar instructor, *Selected Methods of Theoretical Physics*; research mentor, Toda CFT project.

Also: online seminars on quantum field theory for students in Kazakhstan, and physics Olympiad training. Student jury member, All-Russian High School Physics Olympiad (2018–2019) and Maxwell Olympiad (2016).